NUMERICAL OPTIMIZATION - BOOK CLUB

OPTIMISATION QUANT TEAM

1. Fundamentals of Unonstrained Optimization

The concept of a local minimizer is introduced,

Definition (Weak local minimizer)**.** A point x^* is a *weak local minimizer* if there is a neighborhood [1] $\mathcal{N}$ of x^* such that $f(x^*) \leq f(x), \quad \forall x \in \mathcal{N}$.

On the other hand a *strict* local optimizer is where the inequality above is strict. A local minimizer x^* is called *isolated* if there exists a neighborhood in which x^* is the only local minimizer. Sufficient conditions for a point to be a local minimizer were presented as the following

Theorem 1 (Second-Order Sufficient Conditions)**.** Suppose that $\nabla^2 f$ is continous in an open neighborhood of x^* and that $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite. Then x^* is a strict local minimizer of f

1.1. Questions.

i) *Do we use any of these?* Vplot has some unconstrained, least-squares algorithm.
ii) *What have you discussed previously on the topic of smooth optimization?* Interior point methods for optimizing SIMM delta.
iii) *How is the process of inverting a matrix affected by terrible condition number?* Which algorithms are used for matrix inversion? Is there some bound on iterations dependent on the condition number?

1.2. Solutions.

The exercises chosen for this chapter are 2.1, 2.3, 2.10, 2.11 and 2.12.

1.2.1. *Exercise 2.1.*

Compute i) the gradient and ii) the Hessian of the Rosenbrock [2] function

$$f(x) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$$

and iii) show that $x^* = (1, 1)$ is the only local minimizer.

i) The gradient of a function is the vector of all its partial derivatives

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2} \right)^T.$$

Computing the two derivatives in x_1 and x_2 we get

$$\frac{\partial f}{\partial x_1} = -400(x_2 - x_1^2)x_1 - 2(1 - x_1)$$
$$\frac{\partial f}{\partial x_2} = 200(x_2 - x_1^2)$$

ii) The Hessian of a function is the matrix of all it's partial derivatives

$$\mathbf{H}_f = \left(\nabla^2 f\right)_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$$

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[1] A neighborhood of a point x, sometimes denoted $\mathcal{N}_x$ is an open set containing x. Defining which subsets of are considered open is the act of choosing a topology for your space. In this book they use the topology induced on $\mathbb{R}^n$ by the euclidean metric $(\|\cdot\|_2) : x \mapsto \sqrt{x^T x}$. For more on Euclidean topology, see here.

[2] There are higher dimensional generalizations of the Rosenbrock function which are classic problems to benchmark performance on n-dimensional, non-convex probelms.

As a mnemonic, $\nabla = \left(\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_2}\right)$ and $\nabla^2 = \nabla\nabla^T$.

$$\begin{aligned}\frac{\partial^2 f}{\partial x_1} &= -400\left(x_2 - 3x_1^3\right) + 2\\ \frac{\partial^2 f}{\partial x_1 \partial x_2} = \frac{\partial^2 f}{\partial x_2 \partial x_1} &= -400x_1\\ \frac{\partial^2 f}{\partial x_2} &= 200\end{aligned}$$

iii) For a local minimizer, $\nabla f = 0$ and $\nabla^2 f$ is positive definite. The first condition means to solve for which x_1, x_2 the equations

$$\frac{\partial f}{\partial x_1} = 0, \quad \frac{\partial f}{\partial x_2} = 0$$

hold. The second one is the easiest to start with, it implies the relation $x_2 = x_1^2$ for stationary points (a point for which $\nabla f(x) = 0$ is true) of f. Substituting into the former of the equations above we get

$$2(1 - x_1) = 0 \implies x_1 = 1$$

which in turn means $x_2 = 1$. This means there is only one stationary point and thus, at most one local minimizer. Only if the Hessian is positive definite can the stationary point be a local minimizer. Let's check!

$$\nabla^2 f(1,1) = \begin{pmatrix} 802 & -400 \\ -400 & 200 \end{pmatrix}$$

A symmetric matrix is positive definite if all it's leading principal minors are positive[3]. The principal minors are the determinants of square sub-matrices starting from the upper left corner. The first one is the 1×1 matrix in the top left corner, i.e. $802 > 0$. The second one is

$$\det \nabla^2 f(1,1) = 802 \cdot 200 - 400^2 = 400 > 0$$

□

1.2.2. *Exercise 2.3.* Starting with f_1,

$$\frac{\partial f_1}{\partial x_i} = \frac{\partial a^T x}{\partial x_i} = \frac{\partial}{\partial x_i} \sum_i a_i x_i = a_i.$$

So $\nabla f_1 = a \in \mathbb{R}^n$. Since the gradient is constant and the Hessian consists of second derivatives, it is zero $\nabla^2 f_1 = \mathbf{0}$.

For f_2 we have

$$\begin{aligned}\frac{\partial f_2}{\partial x_k} &= \sum_{i,j} A_{ij} \frac{\partial (x_i x_j)}{\partial x_k}\\ &= \sum_{i,j} A_{ij}(\delta_{ik} x_j + x_i \delta_{jk})\\ &= \sum_j A_{kj} x_j + \sum_i A_{ik} x_i\\ &= 2A_k x\end{aligned}$$

where A_k is the kth column of A and δ_{ij} is the Kronecker delta[4]. The last inequality follows from A being symmetric. Thus

$$\nabla f_2 = (2A_1 x, \ldots, 2A_n x)^T = 2Ax.$$

For the Hessian, we need

$$\frac{\partial^2 f_2}{\partial x_l x_k} = \frac{\partial}{\partial x_l}\left(2\sum_i A_{ik} x_i\right) = 2A_{lk}.$$

which gives

$$\nabla^2 f_2 = 2A$$

[3]See Sylvester's criterion

[4]See Kronecker delta

1.2.3. *Exercise 2.10.*

$$\frac{\partial \tilde{f}}{\partial z_i} = \sum_k \frac{\partial f}{\partial x_k} \frac{\partial x_k}{\partial z_i}$$

To begin with, we compute the second factor in the sum above

$$\frac{\partial x_k}{\partial z_i} = \frac{\partial}{\partial z_i} \left(\sum_j S_{kj} z_j + s_k \right) = S_{ki}.$$

Substituting this for $\frac{\partial x_k}{\partial z_i}$ in the first equation we get

$$\frac{\partial \tilde{f}}{\partial z_i} = \sum_k \frac{\partial f}{\partial x_k} S_{ki} = S_i^T \cdot \nabla f$$

with S_i being the ith column of S. This means[5]

$$\nabla \tilde{f}(z) = \begin{bmatrix} S_1 \nabla f \\ \vdots \\ S_n \nabla f \end{bmatrix} = S^T \nabla f(x)$$

For the Hessian, we need

$$\frac{\partial^2 f}{\partial z_i \partial z_j} = \sum_{k,l} \frac{\partial^2 f}{\partial x_k \partial x_l} \frac{\partial x_k}{\partial z_i} \frac{\partial x_l}{\partial z_j} = \sum_{k,l} \frac{\partial^2 f}{\partial x_k \partial x_l} S_{ki} S_{lj}$$

Note that we could have hade second derivatives of x w.r.t. z, but thankfully the first derivative is constant so we are off the hook! The expression above can be rewritten as:

$$\frac{\partial^2 f}{\partial z_i \partial z_j} = S_i^T \cdot \nabla^2 f \cdot S_j$$

So when assembling the whole Hessian matrix from the partial derivatives, we obtain

$$\nabla \tilde{f}(z) = S^T \nabla f(x) S$$

□

1.2.4. *Exercise 2.11.* Assuming a unit step size, the relation between subsequent steps is

$$x_{k+1} = x_k + p$$

We use this relation to make sure the affine relation between the iterates is preserved using this update rule. To begin with, the relation between the iterates z_k and x_k is

$$x_k = S z_k + s.$$

The step direction at each iteration is computed as

$$p_k = -B_k^{-1} \nabla f(x_k)$$

and

$$\tilde{p}_k = -\tilde{B}_k^{-1} \nabla \tilde{f}(z_k)$$

Given that the initial Hessians approximations are chosen s.t $\tilde{B}_0 = S^T B_0 S$ we can see that

$$\tilde{p}_0 = -(S^T B_0 S)^{-1} S^T \nabla f(x_0) = -S^{-1} B_0^{-1} \nabla f(x_0)$$

which, for z_{k+1}, means

$$\begin{aligned} z_{k+1} =& z_k + \tilde{p}_k = z_k - S^{-1} B_k^{-1} \nabla f(x_k) \\ =& (S^{-1}(x_k - s)) - S^{-1} B_k^{-1} \nabla f(x_k) \end{aligned}$$

Applying the affine transformation $v \mapsto Sv + s$ on both sides

$$\underbrace{x_k + B_k^{-1} \nabla f(x_k)}_{x_{k+1}} = S z_{k+1} + s$$

[5] If S_i had been rows, we would have had $\nabla \tilde{f}(z) = S \nabla f(x)$

So, given $\tilde{B}_k = S^T B_k S$, the above holds. Since we can choose the initial approximations such that this holds, it remains to verify that the rank one and BFGS update rules preserve this relationship. We begin with the rank-one update.

$$B_{k+1} \;=\; B_k \;+\; \frac{(y_k - B_k s_k)(y_k - B_k s_k)^T}{(y_k - B_k s_k)^T s_k}$$

where

$$s_k \;=\; x_{k+1} - x_k, \qquad y_k \;=\; \nabla f(x_{k+1}) - \nabla f(x_k).$$

We know that for $k = 0$, we have $x_k = Sz_k + s$ and $\tilde{B}_0 = S^T B_0 S$. If this update preserves the relation between the Hessian iterates, then the relation between the solution iterates will be preserved and continue to hold through further iterations. The quantities y_k and s_k relate to their ˜counterparts as follows:

$$\tilde{s}_k \;=\; z_{k+1} - z_k = S^{-1}(x_{k+1} - x_k) = S^{-1} s_k, \qquad \tilde{y}_k \;=\; \nabla f(z_{k+1}) - \nabla f(z_k) = S^T y_k.$$

We now show the update satisfies behaves desirably by substituting in the above relations,

$$\begin{aligned}
\tilde{B}_{k+1} &= \tilde{B}_k + \frac{(\tilde{y}_k - \tilde{B}_k \tilde{s}_k)(\tilde{y}_k - \tilde{B}_k \tilde{s}_k)^T}{(\tilde{y}_k - \tilde{B}_k \tilde{s}_k)^T \tilde{s}_k} \\
&= S^T B_k S + \frac{(S^T y_k - S^T B_k S S^{-1} s_k)(S^T y_k - S^T B_k S S^{-1} s_k)^T}{(S^T y_k - S^T B_k S S^{-1} s_k)^T S^{-1} s_k} \\
&= S^T B_k S + \frac{S^T (y_k - B_k s_k)(y_k - B_k s_k)^T S}{(y_k - B_k s_k)^T S S^{-1} s_k} \\
&= S^T \underbrace{\left(B_k + \frac{(y_k - B_k s_k)(y_k - B_k s_k)^T}{(y_k - B_k s_k)^T s_k} \right)}_{B_{k+1}} S
\end{aligned}$$

The procedure for the BFGS update rule is identical. □

1.2.5. *Exercise 2.12.* This question is not entirely explicit in how to setup the problem, so I will make some assumptions to illustrate the effect of scaling. Given some "properly scaled" function $g(x_1, x_2)$ and take f to be the following function

$$f(x_1, x_2) = g(k_1 x_1, k_2 x_2)$$

where $k_1 \gg k_2$ are scale factors to make f poorly scaled. Given the Hessian has full rank, we can always choose a coordinate system in which the mixed partial derivatives are 0, so we will simply assume that is the case for $\frac{\partial^2}{\partial x_1 \partial x_2}$. This means the Hessian is:

$$\nabla^2 f = \begin{pmatrix} k_1^2 \frac{\partial^2 g}{\partial x_1^2} & 0 \\ 0 & k_2^2 \frac{\partial^2 g}{\partial x_2^2} \end{pmatrix}$$

The condition number of a matrix, $\kappa(A)$, is the quotient between its largest and smallest singular values (which for a square matrix is the absolute value of its eigenvalues):

$$\kappa(A) = \frac{\max_i |\lambda_i|}{\min_i |\lambda_i|}$$

In our case, that becomes:

$$\kappa\left(\nabla^2 f\right) = \frac{\left| k_1^2 \frac{\partial^2 g}{\partial x_1^2} \right|}{\left| k_2^2 \frac{\partial^2 g}{\partial x_2^2} \right|}$$

Now if we assume that "properly scaled" means that $|\frac{\partial^2 g}{\partial x_1^2}| \approx |\frac{\partial^2 g}{\partial x_2^2}|$ at the optimum then that means

$$\kappa\left(\nabla^2 f\right) \approx \frac{k_1^2}{k_2^2}$$

So if one axis has a characteristic length of $\sim 10^6$ times larger than the other, the condition number will be 10^{12}!

2. Theory of Constrained Optimization